$k: 9, a: 1, b = 7$

Dale

5.5.1. A.

$$L_1 = \lim_{x \to 9} \frac{x^2 - 81}{x - 9} = \frac{(x+9)\cancel{(x-9)}}{\cancel{x-9}} = x + 9$$
$$= 9 + 9 = 18$$

$$L_2 = \lim_{x \to 0} \frac{\sqrt{x + 10^2} - 10}{x} \cdot \frac{\sqrt{x + 10^2} + 10}{\sqrt{x + 10^2} + 10}$$

$$= \frac{(x + 10^2) - 100}{x(\sqrt{x + 10^2} + 10} = \frac{\cancel{x}}{\cancel{x}(\sqrt{x + 100} + 10} = \frac{1}{\sqrt{x + 100} + 10}$$

$$= \frac{1}{\sqrt{0 + 100} + 10} = \frac{1}{10 + 10} = \frac{1}{20} = 0{,}05$$

$$L_3 = \lim_{x \to \infty} = \frac{3x^3 + 7x + 7}{x^3 - x + 9} = \frac{\frac{3x^3}{x^3} + \frac{7x}{x^3} + \frac{7}{x^3}}{\frac{x^3}{x^3} - \frac{x}{x^3} + \frac{9}{x^3}} = \frac{3 + 0 + 0}{1 - 0 + 0}$$
$$= 3$$

5.2 $$L_4 = \lim_{x \to 0} \frac{\sin((1+1)x)\tan((7+2)x)}{x^2} = \frac{\sin(2x)}{x} \cdot \frac{\tan(9x)}{x}$$

$$= \left(\lim_{x \to 0} \frac{\sin(2x)}{x}\right)\left(\lim_{x \to 0} \frac{\tan(9x)}{x}\right) = \frac{2}{1} \cdot \frac{9}{1} = 18$$

5.3. $$R(n) = \frac{100n + 9}{n + 9} \quad \Big| \quad \lim_{n \to \infty} \frac{100n + 9}{n + 9} = \frac{100 + \frac{9}{n}}{1 + \frac{9}{n}}$$
$$= \frac{100 + 0}{1 + 0} = 100$$

pendapatan perbulan untuk jangka sangat panjang sebanyak Rp 100, meskipun dianggap ~~bahwa~~ bervariasi pada akhirnya akan mengantuh titik Rp 100

No

Date

e.

$$\lim_{n\to\infty} \sqrt{n^2+9n} - \sqrt{n^2+7n} \cdot \frac{\sqrt{n^2+9n}+\sqrt{n^2+7n}}{\sqrt{n^2+9n}+\sqrt{n^2+7n}} = \frac{(n^2+9n)-(n^2+7n)}{\sqrt{n^2+9n}+\sqrt{n^2+7n}}$$

$$= \frac{2n}{\sqrt{n^2+9n}+\sqrt{n^2+7n}} = \frac{\frac{2n}{n}}{\sqrt{\frac{n^2}{n^2}+\frac{9n}{n^2}}+\sqrt{\frac{n^2}{n^2}+\frac{7n}{n^2}}} = \frac{2}{\sqrt{1+\frac{9}{n}}+\sqrt{1+\frac{7}{n}}}$$

$$= \frac{2}{\sqrt{1+0}+\sqrt{1+0}} = \frac{2}{1+1} = \frac{2}{2} = 1$$

karena kedua persamaan tsb menilik: dua kelompok pelanggan yang hampir identik.

5.a.

$$f(x) = \begin{cases} (x^2-81)/(x-9), & \text{jika } x < k \\ p & x = k \\ x+9 & x > k \end{cases}$$

$$\lim_{x\to 9^-} = \frac{x^2-81}{x-9} = \frac{(x+9)\cancel{(x-9)}}{\cancel{x-9}} = x+9 = 9+9 = 18$$

$p = 18$

$$\lim_{x\to 9^+} x+9 = 9+9 = 18$$

$$-9 = 18-9 = 9$$

JOYKO 30 Lines, 6 mm